

Limits of Honest Adaptation in a Binary Instrumental-Variable Model

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Abstract

We study honest confidence sets for an overlap-weighted complier effect in a binary monotone instrumental-variable model. The full scalar covariate is observed and has a known uniform distribution, while the instrument propensity is unknown. Nuisance means have known cellwise Hölder smoothness below one quarter relative to a fixed two-cell summary, and the identifying denominator lies in a known stratum that may weaken with sample size. We establish finite-sample lower and upper bounds for expected confidence-set diameter at fixed coverage error. The lower bound holds at an explicit law with constant nuisance means, and a projection-and-inversion confidence set attains the same rate uniformly over the large model. At the same law, an oracle told only that the nuisance means depend on the summary achieves a faster rate. Thus, along some weakening-identification paths, the oracle confidence set contracts while no procedure honest over the large model can have contracting expected diameter at that simple law. Both procedures use the same observations. The proof combines instrumental-variable-compatible alternatives with established testing and projection methods. The result isolates the inferential cost of an unverified dimension-reduction restriction. It is a model-specific application, and rate matching does not ensure practically short confidence sets.

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Keywords: Honest confidence sets; adaptation; instrumental variables; weak identification; expected diameter; nonparametric inference.

1 Introduction and Related Literature

1.1 The question: true simplicity versus justified simplicity

If the conditional means needed for instrumental-variable estimation depend only on a coarse summary of the covariates, imposing that restriction can simplify inference. The restriction may, however, be uncertain. Can a confidence procedure exploit simple nuisance functions when they are present while remaining valid over a larger class in which those functions vary within the summary cells? This question concerns the information needed to justify a short confidence set, not merely the accuracy of an estimated conditional mean.

We study this distinction in a deliberately restricted binary instrumental-variable model. The outcome Y , treatment D , and instrument Z are binary; the full covariate X is observed and has a known uniform distribution on $[0, 1]$. The instrument propensity is unknown. The large model consists of valid conditional-independence, exclusion, and treatment-monotonicity laws, with overlap and observed-cell interiority restrictions. Its three nuisance means, $E[Z | X]$, $E[D | X]$, and $E[Y | X]$, have known fixed Hölder smoothness $\beta \in (0, 1/4)$ within the two cells of $S(X) = \mathbf{1}\{X \geq 1/2\}$. The simple submodel restricts those means to be functions of S . It does not require the entire conditional joint law to depend on X only through S .

Our target is

$$\tau(P) = \frac{N(P)}{\Lambda(P)}, \quad N(P) = E_P[\text{Cov}_P(Z, Y | X)], \quad \Lambda(P) = E_P[\text{Cov}_P(Z, D | X)]. \quad (1.1)$$

Under the IV restrictions, this is an overlap-weighted complier effect, with range $[-1, 1]$, rather than the unweighted local average treatment effect. We work on known identification strata $\lambda_n \leq$

$\Lambda(P) \leq 2\lambda_n$, where $0 < \lambda_n \leq 1/32$ and the actual denominator remains unknown. The strength of identification may decrease with sample size, while overlap and cell-interiority bounds remain fixed. The precise model and information available to procedures are specified in Section 2.

1.2 A same-law comparison and its contribution

Theorem 3.1 identifies the optimal expected diameter of confidence sets that are honest over the large model, even when diameter is evaluated at one explicit simple law. At that law, P_n^* , all three nuisance means equal $1/2$. Nevertheless, every full-model-honest procedure has expected diameter bounded below by a positive multiple of

$$q_{\beta,n} = \min \left\{ 1, \frac{n^{-4\beta/(4\beta+1)}}{\lambda_n} \right\}. \quad \text{The simple-model oracle rate is } q_{0,n} = \min \left\{ 1, \frac{n^{-1/2}}{\lambda_n} \right\}. \quad (1.2)$$

A single projection-based confidence procedure attains the first rate uniformly over the entire large model. Proposition 3.2 establishes the second rate under honesty restricted to the simple model, with its lower bound evaluated at the same P_n^* . The oracle knows that the simple restriction holds, not the nuisance values. Both procedures observe the full data. The reference law may vary with n through λ_n ; “same law” means the same comparison point at each sample size and identification stratum.

These are finite-sample lower and upper bounds, up to constants, for every $n \geq 2$ and fixed coverage error $\alpha \in (0, 1/2)$. Confidence sets may be randomized, empty, or disconnected; their diameter is measured after intersection with the true parameter range. In particular, when

$$n^{-1/2} \ll \lambda_n \lesssim n^{-4\beta/(4\beta+1)}, \quad (1.3)$$

the oracle’s optimal expected diameter contracts, whereas no full-model-honest procedure has contracting expected diameter at the specified simple law. For $\beta = 1/8$, the upper scale in (1.3) is $n^{-1/3}$. This separation is not caused by withholding X from the procedure or by the failure of a

particular fitting algorithm.

The contribution is a model-specific application theorem connecting three features: alternatives that retain a genuine monotone-IV representation, a lower bound at an explicit simple law shared with the oracle comparison, and a matching upper bound on the same large model. The causal feasibility of the alternatives is substantive: in our construction the perturbation must satisfy $h^2 \leq c$, while the first-stage moment remains $c/4$. One cannot obtain this result solely by quoting a worst-case lower bound from a larger unconstrained covariance model and dividing by a small denominator. At the same time, the underlying testing, projection, and inversion methods are established tools. We do not claim a new general principle of adaptation or a new estimation algorithm.

1.3 Relation to existing results

The distinction between the model on which coverage is required and the model on which length is evaluated is central to Cai and Low (2004). Their Theorem 1 bounds expected length using a between-class modulus in Gaussian experiments with linear functionals. Pointwise obstacles at simple distributions are also not new: Cai and Guo (2015, version 2), Theorem 3 and Corollary 1, establish nonadaptivity at specified low-complexity distributions in high-dimensional Gaussian regression. Their Lemma 1 gives a two-target coverage-transfer argument. Our comparison applies these ideas to a nonlinear causal ratio in a different experiment; the use of a simple reference distribution is not itself a methodological innovation.

The nonparametric rate in (1.2) likewise has an established origin. Robins et al. (2009), Section 4 and Theorem 4.1, study minimax estimation of expected conditional covariance under Hölder restrictions. Their general mixture bound, Theorem 2.1, includes the single-null specialization described immediately after the theorem. Thus an explicit simple null does not put our lower-bound argument beyond their framework. Section 4 summarizes the full-observation likelihood calculation, and Supplement Section S.7 shows that the IV alternatives can also be generated from a binary-regression source by a transformation independent of the unknown perturbation. This is

a one-way transfer of a testing difficulty, not an increase in information through synthetic instruments.

Higher-order estimation and low-regularity effect inference are also precedents, not new components of this paper. Li et al. (2011), Sections 1, 3, and 6, construct projection-based U-statistics for expected conditional covariance and formulate confidence sets for a variance-weighted treatment effect by inverting covariance inference for a transformed outcome. Their treatment-variance denominator and causal interpretation differ from our IV denominator and complier target. Our upper bound is a deliberately elementary use of this projection-and-inversion strategy, with a deterministic bias allowance and a separate expected-diameter argument. Its value here is to close the lower-bound comparison on the stated model, not to introduce low-regularity confidence sets for the first time.

Cross-fitting provides another related route to covariance estimation. Newey and Robins (2018, version 1) analyze fast remainders for cross-fitted semiparametric estimators; their partially linear projection result, Theorem 5, assumes a nonsingular treatment-variance matrix. McClean et al. (2025, version 3), Theorems 2 and 3, obtain covariance estimation rates and a slower-than-root- n central limit theorem using known-density, undersmoothed kernel estimators. The latter theorem adds boundedness and conditional-variance continuity to its preceding conditions. These results concern different procedures and guarantees. We neither infer full-model finite-sample honesty from a point-estimation rate nor claim that our conservative construction improves their performance.

For weakly identified ratios, Noack and Rothe (2023, version 4), Section 3 and Theorem 1, develop bias-aware Anderson–Rubin confidence sets in fuzzy regression discontinuity designs. Their uniform asymptotic coverage result uses their regression-smoothness, design, bandwidth, and standard-error conditions. In a conditional-IV setting with high-dimensional covariates, Ma (2025, version 5), Theorem 3.2 under Assumptions 3.1–3.2, establishes uniform asymptotic size for an orthogonalized Anderson–Rubin test for the LATE. Those assumptions include sparse nuisance specifications and estimation restrictions. These works establish that bias-aware inversion

and identification-robust IV inference are existing methods. Our additional question is the optimal expected diameter at a specified simple law under a rougher, fixed Hölder model; their assumptions and targets are not silently substituted for ours.

Noncontraction under weak identification is also an established limitation. Smucler et al. (2025, version 3), Corollary 2, gives a worst-case diameter obstruction to uniform asymptotic coverage for integral-equation functionals under its model and regularity conditions; its examples include IV effects. Our question is more restricted: on the specified Hölder and identification strata, what diameter is necessary at the same simple law at which a small-model oracle can contract? An impossibility result over a broader coverage model cannot simply be restricted to a smaller one. The IV-compatible alternatives, the oracle benchmark at P_n^* , and attainment on the same $\mathcal{P}_{1,n}(\beta)$ require the arguments below; we do not claim a new general weak-identification impossibility principle.

Babii (2020, version 4), Theorems 3.1 and 3.2 under Assumptions 3.1–3.2, establishes coverage and size guarantees for regularized NPIV structural-function bands. Honest IV inference and analysis of set size therefore predate this paper. His source-condition index is not the Hölder index of our three nuisance means. We study a scalar overlap-weighted complier effect and the cost of requiring honesty beyond a simple nuisance submodel, not a band for the structural function in an ill-posed NPIV model. Our rate comparison does not follow by identifying these different smoothness parameters.

Taken together, these connections place the result as a restricted application of established theory. The specific IV construction and same-model comparison provide the content of the application. Differences in notation, the use of an explicit null, or finite-sample rather than asymptotic wording would not, by themselves, establish that contribution.

1.4 Scope and organization

The result concerns adaptation to nuisance simplicity within a known smoothness class, not adaptation to unknown β . It assumes a known covariate distribution and a fixed two-cell summary;

it does not establish validity for a learned representation, unknown design distribution, higher-dimensional model, or $\beta \geq 1/4$. The target remains (1.1). The upper bound is rate matching, not constant-sharp or a promise of practically short confidence sets. Section 5 states this limitation explicitly; the supplementary analytic exhibits do not claim simulated coverage or estimator dominance.

Section 2 defines the experiment, target, models, and risk. Section 3 states the main result and oracle benchmark. Section 4 gives the proof strategy and attaining procedure, Section 5 interprets the regimes and finite-sample limitations, and Section 6 concludes. The supplement contains the complete proofs and additional analytic exhibits.

2 Model, Target, and Honest Inference

2.1 Observations and instrumental-variable restrictions

We observe $W^n = (W_1, \dots, W_n)$, where $W_i = (Y_i, D_i, Z_i, X_i)$ are independent draws from a law P , $n \geq 2$, and $Y, D, Z \in \{0, 1\}$. The covariate distribution is known: $X \sim U[0, 1]$. The fixed summary is $S(X) = \mathbf{1}\{X \geq 1/2\}$. Every procedure may use the full observations, including X ; the experiment does not restrict the data to (Y, D, Z, S) . Neither the summary nor its two cells is selected from the data.

Write

$$\zeta(x) = P(Z = 1 \mid X = x), \quad r_{dy|z}(x) = P(D = d, Y = y \mid Z = z, X = x), \quad d, y, z \in \{0, 1\}. \quad (2.1)$$

The distribution of X is known, but the propensity function ζ is unknown. Throughout, the conditional cell probabilities sum to one for each z , and, for almost every x ,

$$\frac{1}{4} \leq \zeta(x) \leq \frac{3}{4}, \quad r_{dy|z}(x) \geq \kappa := \frac{1}{16}, \quad r_{1y|1}(x) \geq r_{1y|0}(x), \quad r_{0y|0}(x) \geq r_{0y|1}(x). \quad (2.2)$$

The last two inequalities are imposed for both outcome values. They encode more than a non-negative first stage. Specifically, our causal model admits binary potential treatments $D(0), D(1)$ and potential outcomes $Y(0), Y(1)$ satisfying

$$(Y(0), Y(1), D(0), D(1)) \perp Z \mid X, \quad D(1) \geq D(0), \quad D = D(Z), \quad Y = Y(D). \quad (2.3)$$

Writing the outcome as $Y(D)$ incorporates exclusion of a direct instrument effect. The model allows dependence between the potential outcomes and treatment type, and does not require unconditional independence from X .

For an observed law satisfying the overlap and cell-interiority bounds, the two families of monotonicity inequalities in (2.2) are equivalent to the existence of a representation satisfying (2.3). Suppressing x , define the conditional probabilities of always-takers, never-takers, and compliers by

$$p_{AT} = \sum_y r_{1y|0}, \quad p_{NT} = \sum_y r_{0y|1}, \quad p_{CP} = 1 - p_{AT} - p_{NT}. \quad (2.4)$$

The two sums of nonnegative cell differences in (2.2) both equal p_{CP} . Let T denote treatment type and write $q_{T,d}(x) = P(Y(d) = 1 \mid T, X = x)$. The outcome success probabilities needed to reproduce the observed cells are

$$\begin{aligned} q_{AT,1} &= \frac{r_{11|0}}{p_{AT}}, & q_{NT,0} &= \frac{r_{01|1}}{p_{NT}}, \\ q_{CP,1} &= \frac{r_{11|1} - r_{11|0}}{p_{CP}}, & q_{CP,0} &= \frac{r_{01|0} - r_{01|1}}{p_{CP}}. \end{aligned} \quad (2.5)$$

Cell interiority makes p_{AT} and p_{NT} positive. When $p_{CP} = 0$, all complier cell differences van-

ish, and its outcome probabilities may be set to $1/2$. Otherwise, the ratios in (2.5) belong to $[0, 1]$. Set the remaining, unobserved potential-outcome margins to Bernoulli($1/2$) and, for example, use a product coupling within each type. Define $D(0) = \mathbf{1}\{T = \text{AT}\}$ and $D(1) = \mathbf{1}\{T \in \{\text{AT}, \text{CP}\}\}$. Drawing Z with success probability $\zeta(X)$ independently of these latent variables conditional on X , and using $D = D(Z)$ and $Y = Y(D)$, reproduces the given observed law. The product coupling establishes existence; independence between the potential outcomes is not imposed on the model. Conversely, under (2.3), each cell difference in (2.2) is the probability of a complier with the corresponding potential outcome and is nonnegative.

2.2 Target parameter and identification strength

Let $\pi(x) = E_P[D \mid X = x]$ and $\rho(x) = E_P[Y \mid X = x]$. Define

$$\begin{aligned} N(P) &= E_P[ZY] - E_P[\zeta(X)\rho(X)] = E_P[\text{Cov}_P(Z, Y \mid X)], \\ \Lambda(P) &= E_P[ZD] - E_P[\zeta(X)\pi(X)] = E_P[\text{Cov}_P(Z, D \mid X)], \\ \tau(P) &= \frac{N(P)}{\Lambda(P)}. \end{aligned} \tag{2.6}$$

To interpret the ratio, put $\theta_{\text{CP}}(x) = q_{\text{CP},1}(x) - q_{\text{CP},0}(x)$, with any value in $[-1, 1]$ when $p_{\text{CP}}(x) = 0$. The representation above gives

$$\begin{aligned} w_P(x) &= \zeta(x)\{1 - \zeta(x)\}p_{\text{CP}}(x), \\ \Lambda(P) &= E_P[w_P(X)], \quad N(P) = E_P[w_P(X)\theta_{\text{CP}}(X)]. \end{aligned} \tag{2.7}$$

Thus $\tau(P)$ is an overlap-weighted complier effect. It is not, in general, the unweighted average effect among compliers, the average treatment effect, or a homogeneous structural coefficient. Nonnegative weights and binary outcomes imply that $\tau(P) \in \Theta := [-1, 1]$ whenever $\Lambda(P) > 0$.

For each sample size, fix a known identification scale λ_n and restrict the law by

$$0 < \lambda_n \leq \frac{1}{32}, \quad \lambda_n \leq \Lambda(P) \leq 2\lambda_n. \tag{2.8}$$

Knowing this stratum does not reveal $\Lambda(P)$ or the nuisance functions. The sequence λ_n may tend to zero; no positive lower bound on $p_{\text{CP}}(x)$ is imposed. In contrast, the observed-cell and instrument-overlap bounds in (2.2) remain fixed. These are substantive restrictions on the model, not assertions covering every weak-instrument experiment.

2.3 The large model and its simple submodel

Fix a known smoothness index $\beta \in (0, 1/4)$. For cell $s \in \{0, 1\}$, use the rescaled coordinate $u = 2x - s$ and let $f_s(u) = f((s + u)/2)$ on the cell interior, with its one-sided continuous extension to $[0, 1]$. Define

$$\mathcal{H}_S^\beta(2) = \left\{ f : \max_{s \in \{0, 1\}} \|f_s\|_\infty \leq 2, \quad \max_{s \in \{0, 1\}} \sup_{\substack{u, v \in [0, 1] \\ u \neq v}} \frac{|f_s(u) - f_s(v)|}{|u - v|^\beta} \leq 2 \right\}. \quad (2.9)$$

No continuity is required across the two cells. Their extensions may have different limiting values at $x = 1/2$; a value at that single point does not affect the law or the target. Membership of a conditional-mean function means that a version with these properties exists.

Let $\mathcal{P}_{1,n}(\beta)$ be the set of observed laws with the distribution of X specified above, satisfying (2.2), (2.8), and

$$\zeta, \pi, \rho \in \mathcal{H}_S^\beta(2). \quad (2.10)$$

By Section 2.1, every such law admits a valid monotone-IV representation. Smoothness is imposed on these three observed conditional means, not on every $r_{dy|z}$, on latent outcome means, or on conditional variances. The restrictions on those other quantities are only those implied by the stated assumptions. The distribution of X , the fixed summary, β , the radius, the bounds in (2.2), and λ_n are available to procedures. In particular, a procedure may depend on β ; simultaneous adaptation to unknown smoothness is not the question here.

The simple model is

$$\mathcal{P}_{0,n} = \{P \in \mathcal{P}_{1,n}(\beta) : \zeta, \pi, \rho \text{ are measurable with respect to } S(X)\}. \quad (2.11)$$

Each of the three functions is then constant within each summary cell, up to null sets. Since these constants lie in $[0, 1]$ and their cellwise Hölder seminorms vanish, $\mathcal{P}_{0,n}$ is the same set for every $\beta \in (0, 1/4)$. Its definition does not require the full conditional joint law of (Y, D, Z) to depend on X only through S . Simplicity concerns the conditional means needed in (2.6), not statistical sufficiency of S for the whole experiment. An oracle informed that $P \in \mathcal{P}_{0,n}$ still does not know those means.

2.4 Honest confidence sets and diameter risk

Fix $\alpha \in (0, 1/2)$. An admissible procedure returns a measurable random closed subset $C_n = C_n(W^n, U)$ of Θ , where $U \sim \nu$ is auxiliary randomness independent of the observations and its law does not depend on P . Deterministic procedures are included. Measurability is understood to ensure that coverage events and the diameter are measurable. All probabilities and expectations below include U ; write $\mathbb{P}_{P,n} = P^{\otimes n} \otimes \nu$ and $\mathbb{E}_{P,n}$ for this joint experiment.

For a model $\mathcal{B}$, finite-sample honesty means membership in

$$\mathfrak{C}_{\alpha,n}(\mathcal{B}) = \left\{ C_n : \inf_{P \in \mathcal{B}} \mathbb{P}_{P,n} \{ \tau(P) \in C_n \} \geq 1 - \alpha \right\}. \quad (2.12)$$

The same procedure must satisfy the coverage requirement at every law in $\mathcal{B}$. This is a finite-sample requirement at each n , including when the identification stratum changes with n .

For nonempty $C \subseteq \Theta$, let $\text{diam}(C) = \sup_{t, t' \in C} |t - t'|$, and set $\text{diam}(\emptyset) = 0$. Disconnected sets are allowed: diameter measures their total span, not their Lebesgue measure. An empty set always fails to cover and is not discarded when evaluating either coverage or expected diameter. Because Θ is the actual parameter range, intersecting a closed confidence set in $\mathbb{R}$ with Θ preserves coverage and cannot increase diameter. Our loss always concerns this intersection and lies in $[0, 2]$.

For nonempty nested models $\mathcal{A} \subseteq \mathcal{B}$, define

$$L_{\alpha,n}(\mathcal{A}; \mathcal{B}) = \inf_{C_n \in \mathcal{C}_{\alpha,n}(\mathcal{B})} \sup_{P \in \mathcal{A}} \mathbb{E}_{P,n}[\text{diam}(C_n)]. \quad (2.13)$$

The first argument specifies where diameter is evaluated; the second specifies where coverage is required. The procedure $C_n = \Theta$ is always admissible, so the infimum is over a nonempty class. Our adaptation criterion is $L_{\alpha,n}(\mathcal{P}_{0,n}; \mathcal{P}_{1,n}(\beta))$. It evaluates performance on the simple model while requiring honesty on the large model. The oracle benchmark is $L_{\alpha,n}(\mathcal{P}_{0,n}; \mathcal{P}_{0,n})$, which changes the coverage model, not the observed data or the availability of nuisance values. We also use $L_{\alpha,n}(\{P\}; \mathcal{P}_{1,n}(\beta))$ to evaluate full-model-honest procedures at a specified simple law, and $L_{\alpha,n}(\mathcal{P}_{1,n}(\beta); \mathcal{P}_{1,n}(\beta))$ for the full-model minimax risk. Distinguishing these criteria is essential: a worst-case lower bound over the large model alone does not establish a cost of honesty at a simple law.

3 Main Results and the Oracle Benchmark

Throughout this section, the experiment, models, target, and diameter risk are those of Section 2. In particular, the propensity is unknown, the summary is fixed, and simplicity of the small model concerns only the three nuisance means. Expectations include any auxiliary randomization of the confidence-set procedure.

3.1 A common simple reference law

For each admissible identification scale, put $c_n = 6\lambda_n$ and define $P^* = P_n^*$ by $X \sim U[0, 1]$ and

$$P^*(Z = z, D = d, Y = y \mid X = x) = \frac{1 + c_n(2z - 1)(2d - 1)}{8}, \quad z, d, y \in \{0, 1\}. \quad (3.1)$$

This is the joint conditional distribution of (Z, D, Y) given X , not the distribution of (D, Y) conditional on Z . Its propensity is $1/2$, and its Z -conditional cell probabilities are $\{1 + c_n(2z - 1)(2d - 1)\}/4$. Since $0 < c_n \leq 3/16$, each such cell is at least $13/64$. Each of the four monotonicity

differences in (2.2) equals $c_n/2$. Thus the law admits the IV representation of Section 2.1 and satisfies the required overlap and interiority restrictions. Direct summation gives

$$\zeta = \pi = \rho = \frac{1}{2}, \quad N(P^*) = 0, \quad \Lambda(P^*) = \frac{c_n}{4} = \frac{3}{2}\lambda_n, \quad \tau(P^*) = 0. \quad (3.2)$$

Consequently, $P^* \in \mathcal{P}_{0,n}$ for every $\beta \in (0, 1/4)$. The notation suppresses dependence on n through λ_n ; along a weakening identification sequence, the reference law need not be fixed. It does not depend on β or on the confidence-set procedure. Both the full-model and oracle lower bounds below are evaluated at this same law. Its constant propensity is a property of the reference law, not information supplied to procedures throughout either model.

3.2 Honest adaptation at the simple law

For known, fixed $\beta \in (0, 1/4)$, define

$$r_\beta = \frac{4\beta}{4\beta + 1}, \quad q_{\beta,n} = \min \left\{ 1, \frac{n^{-r_\beta}}{\lambda_n} \right\}. \quad (3.3)$$

The exponent r_β is strictly below $1/2$. We use the following non-sharp constants, which depend only on the fixed coverage error:

$$\underline{k}_\alpha = \frac{2(1-2\alpha)^2}{3465} > 0, \quad \bar{k}_\alpha = 16 \left(4 + 2\sqrt{42/\alpha} \right). \quad (3.4)$$

Theorem 3.1 (Honest adaptation and matching diameter). Fix $\alpha \in (0, 1/2)$ and a known $\beta \in (0, 1/4)$. For every $n \geq 2$ and every $0 < \lambda_n \leq 1/32$, the following statements hold under the model restrictions of Section 2.

(i) **Lower bound at the same simple law.** Every $C_n \in \mathfrak{C}_{\alpha,n}(\mathcal{P}_{1,n}(\beta))$ satisfies

$$\mathbb{E}_{P^*,n}[\text{diam}(C_n)] \geq \underline{k}_\alpha q_{\beta,n}. \quad (3.5)$$

The bound includes procedures using the full covariates, auxiliary randomization, and possibly

empty or disconnected closed sets, with the conventions of Section 2.4.

(ii) Attainment on the entire large model. There exists a confidence-set procedure C_n^β , the same for every law in $\mathcal{P}_{1,n}(\beta)$, such that

$$\begin{aligned} \inf_{P \in \mathcal{P}_{1,n}(\beta)} \mathbb{P}_{P,n} \{ \tau(P) \in C_n^\beta \} &\geq 1 - \alpha, \\ \sup_{P \in \mathcal{P}_{1,n}(\beta)} \mathbb{E}_{P,n} [\text{diam}(C_n^\beta)] &\leq \bar{k}_\alpha q_{\beta,n}. \end{aligned} \tag{3.6}$$

The procedure may use n, α, β and the known model specifications of Section 2, but not unknown features of P .

In particular, the same constants bound three distinct risks:

$$\begin{aligned} \underline{k}_\alpha q_{\beta,n} &\leq L_{\alpha,n}(\{P^*\}; \mathcal{P}_{1,n}(\beta)) \\ &\leq L_{\alpha,n}(\mathcal{P}_{0,n}; \mathcal{P}_{1,n}(\beta)) \\ &\leq L_{\alpha,n}(\mathcal{P}_{1,n}(\beta); \mathcal{P}_{1,n}(\beta)) \leq \bar{k}_\alpha q_{\beta,n}. \end{aligned} \tag{3.7}$$

The middle inequalities follow from nesting of the sets on which diameter is evaluated; the coverage model remains $\mathcal{P}_{1,n}(\beta)$ in every term. Hence all three risks have order $q_{\beta,n}$. The first bound is stronger than a worst-case lower bound over the large model: the cost is already present at the specified law with constant nuisance functions. It does not assert a pointwise lower bound of this size at every law in $\mathcal{P}_{0,n}$.

The proof has two parts. The lower-bound argument constructs valid IV alternatives around (3.1) and transfers their coverage requirements to P^* . The upper-bound argument uses projection estimates of the two moments with a deterministic bias allowance, followed by inversion. Section 4 summarizes both arguments, with complete proofs in Supplement Sections S.1–S.2. For the upper bound in (3.6), the resolution is fixed before seeing the data at $J = \lceil n^{2/(4\beta+1)} \rceil$ bins per summary cell. The construction and the control of expected diameter, including the event of an inaccurate denominator estimate, are summarized in Section 4.2 and proved in Supplement Section S.2.

Although the constants in (3.4) can be chosen independently of β throughout the stated inter-

val, the attaining procedure's resolution and bias allowance depend on β . The theorem therefore concerns adaptation to the simple submodel within a specified smoothness class, not adaptation to unknown smoothness. All bounds are finite-sample bounds over the displayed ranges of n and λ_n ; no joint limit of α with n is asserted.

3.3 The oracle benchmark at the same law

Put

$$q_{0,n} = \min \left\{ 1, \frac{n^{-1/2}}{\lambda_n} \right\}. \quad (3.8)$$

Here the subscript 0 denotes the simple-model benchmark, not a smoothness value $\beta = 0$.

Proposition 3.2 (Simple-model oracle). For fixed $\alpha \in (0, 1/2)$ there are constants $0 < \underline{k}_{0,\alpha} \leq \bar{k}_{0,\alpha} < \infty$, depending only on α , such that for all $n \geq 2$ and $0 < \lambda_n \leq 1/32$,

$$\begin{aligned} \underline{k}_{0,\alpha} q_{0,n} &\leq L_{\alpha,n}(\{P^*\}; \mathcal{P}_{0,n}) \\ &\leq L_{\alpha,n}(\mathcal{P}_{0,n}; \mathcal{P}_{0,n}) \leq \bar{k}_{0,\alpha} q_{0,n}. \end{aligned} \quad (3.9)$$

The lower bound holds for every procedure honest on $\mathcal{P}_{0,n}$, evaluated at the same P^* as in Theorem 3.1. The upper bound is achieved by a procedure honest on all of $\mathcal{P}_{0,n}$, without knowing the values of ζ, π, ρ or $\Lambda(P)$.

The oracle information is only that the simple-model restriction is correct. On that model the two-cell projection has zero moment bias, even if the full conditional joint law remains more complicated than a function of S . Its moment errors can be controlled at the $n^{-1/2}$ scale. A two-point family inside the simple model, with one endpoint equal to (3.1), supplies the matching lower bound. The corresponding lower- and upper-bound arguments are summarized in Section 4.3 and proved in Supplement Sections S.1.5 and S.2.5. No known-propensity experiment or extra sample is introduced in this comparison.

3.4 Contraction and its limitations

Fix α and β as above, and consider any sequence $0 < \lambda_n \leq 1/32$. Each full-model-honest optimal risk in (3.7) tends to zero if and only if

$$\lambda_n n^{r_\beta} \longrightarrow \infty. \quad (3.10)$$

The two oracle risks in (3.9) tend to zero if and only if $\lambda_n \sqrt{n} \rightarrow \infty$. These are statements about optimal achievable risks, not a claim that every honest procedure contracts under those conditions. Because $r_\beta < 1/2$, there is a nonempty range

$$n^{-1/2} \ll \lambda_n \lesssim n^{-r_\beta} \quad (3.11)$$

in which oracle expected diameter contracts while every full-model-honest procedure has expected diameter bounded away from zero at P^* . Here $a_n \ll b_n$ means $a_n/b_n \rightarrow 0$, and $a_n \lesssim b_n$ means that this ratio is eventually bounded by a fixed finite constant. For any sequence of such honest procedures in (3.11), diameter also cannot converge to zero in $\mathbb{P}_{P^*,n}$ -probability: a random variable bounded between zero and two that converges to zero in probability must have expectation tending to zero. This does not require a confidence set to equal the entire parameter space, nor does it give a lower bound on its diameter for every realized sample.

For the special value $\beta = 1/8$, $r_\beta = 1/3$. The full-model rate becomes $\min\{1, n^{-1/3}/\lambda_n\}$, while the oracle rate remains $\min\{1, n^{-1/2}/\lambda_n\}$. Thus (3.11) specializes to $n^{-1/2} \ll \lambda_n \lesssim n^{-1/3}$. The boundary $\beta = 1/4$, smoother classes, unknown covariate distributions, learned summaries, and unknown-smoothness adaptation are outside the theorem.

Finally, matching rates do not imply sharp constants or short confidence sets at available sample sizes. The diameter is at most two, so (3.6) may yield only the trivial upper bound after taking its minimum with two. The constants in (3.4) are sufficient for the rate statement, not an assertion of practical efficiency. The target throughout remains the overlap-weighted complier effect from (2.6); no conclusion for an unweighted LATE is obtained by changing its label.

4 Proof Strategy and Attaining Procedure

This section records the constructions and inequalities that determine the rates in Theorem 3.1 and Proposition 3.2. Complete finite-sample proofs, including the likelihood algebra, integer rounding, inversion geometry, and measurability details, are in Supplement Sections S.1–S.2 and S.4–S.12.

4.1 The full-model lower bound

Fix $c = 6\lambda_n$ and the reference law P^* in (3.1). For binary z, d, y , write $\sigma_z = 2z - 1$ and analogously define σ_d, σ_y . Given a measurable h , keep $X \sim U[0, 1]$ and set

$$p_h(z, d, y \mid x) = \frac{1 + c\sigma_z\sigma_d + h(x)(\sigma_z + \sigma_y)}{8}.$$

If $\|h\|_\infty \leq 1/4$ and $h^2 \leq c$, the eight cells are interior and the four treatment-monotonicity differences are nonnegative. The reconstruction in Supplement Section S.4 then supplies potential treatments and outcomes satisfying conditional instrument independence, exclusion, and monotonicity. Direct summation gives

$$\zeta_h = \rho_h = (1 + h)/2, \quad \pi_h = 1/2, \quad \Lambda(P_h) = c/4, \quad N(P_h) = -\frac{1}{4} \int_0^1 h(x)^2 dx.$$

Let $\varphi(u) = 16u^2(1 - u)^2(2u - 1)$ on $[0, 1]$, extended by zero, and let $\mu_2 = \int \varphi^2 = 128/3465$. Partition each of the two summary cells into J blocks and put $h_v = \delta v_j \varphi$ on block j , using one sign vector v for the entire sample. With

$$J = \left\lceil n^{2/(4\beta+1)} \right\rceil, \quad \delta^2 = \frac{1 - 2\alpha}{4} \min\{c, J^{-2\beta}\},$$

all component laws belong to $\mathcal{P}_{1,n}(\beta)$, retain $\Lambda = c/4$, and have the common target $-\Delta$, where $\Delta = \mu_2 \delta^2 / c$. The constraint $\delta^2 \leq c$ is the step that keeps the perturbations inside the monotone-IV model.

For the shared-sign sample mixture $Q_n = 2^{-2J} \sum_v P_v^{\otimes n}$, the exact likelihood calculation in Supplement Section S.1.3 yields

$$1 + \chi^2(Q_n, (P^*)^{\otimes n}) \leq \exp \left\{ \frac{n^2 \xi_c^2 \delta^4}{4J} \right\}, \quad \xi_c = \frac{2\mu_2}{1-c^2} < 1,$$

and hence $\text{TV}(Q_n, (P^*)^{\otimes n}) \leq (1 - 2\alpha)/4$. Averaging coverage over the components, transferring it to P^* , and requiring joint inclusion of 0 and $-\Delta$ gives

$$E_{P^*,n}[\text{diam}(C_n)] \geq \frac{2(1-2\alpha)^2}{3465} \min \left\{ 1, \frac{n^{-4\beta/(4\beta+1)}}{\lambda_n} \right\}.$$

This is a full-observation argument: the distance includes X and D , the mixture is not treated as a causal law, and no restriction to procedures using only S is imposed.

4.2 The full-model upper bound

For a fixed integer J , let K_J be the known-design projection kernel for the $2J$ equal bins of X . For $V \in \{D, Y\}$ define the off-diagonal projection estimate

$$\widehat{M}_{V,J} = \frac{1}{n} \sum_{i=1}^n Z_i V_i - \frac{1}{n(n-1)} \sum_{i \neq \ell} Z_i K_J(X_i, X_\ell) V_\ell.$$

Its expectation differs from $M_V = E[\text{Cov}(Z, V | X)]$ by the inner product of the two projection residuals. Cellwise Hölder approximation and the full U-statistic decomposition imply, uniformly over the large model,

$$|E\widehat{M}_{V,J} - M_V| \leq J^{-2\beta}, \quad \text{Var}(\widehat{M}_{V,J}) \leq \frac{1}{n} + \frac{2(1+2J)}{n(n-1)}.$$

These inequalities remain valid when $J > n$. Let $b_J = J^{-2\beta}$ and let $R_{n,J}$ be the sum of b_J and a dyadic Chebyshev radius for the displayed variance bound. With $\widehat{N}_J = \widehat{M}_{Y,J}$ and $\widehat{\Lambda}_J = \widehat{M}_{D,J}$, invert the two moment inequalities over the known parameter range:

$$C_{n,J} = \left\{ t \in [-1, 1] : |\hat{N}_J - t\hat{\Lambda}_J| \leq (1 + |t|)R_{n,J} \right\}.$$

At $J = \lceil n^{2/(4\beta+1)} \rceil$, simultaneous moment control gives finite-sample coverage at least $1 - \alpha$.

A separate calculation on the event $|\hat{\Lambda}_J - \Lambda| \leq R_{n,J}$ and its complement gives

$$\sup_{P \in \mathcal{P}_{1,n}(\beta)} E_{P,n}[\text{diam}(C_{n,J})] \leq 16 \left(4 + 2\sqrt{42/\alpha} \right) \min \left\{ 1, \frac{n^{-4\beta/(4\beta+1)}}{\lambda_n} \right\}.$$

Supplement Section S.2 proves the projection bias, the variance bound including its degenerate component, exact inversion for possibly disconnected sets, and the expected-diameter inequality. The procedure uses β but neither the unknown law nor the realized denominator.

4.3 The simple-model oracle

For the oracle lower bound, a one-dimensional latent IV family keeps all three nuisance means constant, has one endpoint equal to P^* , and changes τ by order $\min\{1, n^{-1/2}/\lambda_n\}$. Its sample distance is controlled directly, so the same two-target argument applies at the same reference law. For the upper bound, S -measurability makes the two-cell projection bias exactly zero. Taking $J = 1$ in the preceding construction gives the matching $n^{-1/2}/\lambda_n$ scale without revealing the nuisance values. The complete arguments are in Supplement Sections S.1.5 and S.2.5.

5 Implications and Finite-Sample Limitations

5.1 Contraction regimes

Write

$$\lambda_n = n^{-a}, \quad r_\beta = \frac{4\beta}{4\beta+1} \in (0, 1/2). \quad (6.1)$$

Theorem 3.1 and Proposition 3.2 imply three regimes: both optimal expected-diameter risks

contract when $a < r_\beta$; only the simple-model oracle risk contracts when $r_\beta \leq a < 1/2$; and neither contracts when $a \geq 1/2$. Equality belongs to the noncontracting side. For $\beta = 1/8$, the oracle-only window is $1/3 \leq a < 1/2$.

The phrase “permits contraction” concerns the infimum risk, not every honest procedure. In the oracle-only region, full-model honesty prevents contraction at P_n^* even though its nuisance means are constant. The reference law may vary with n through λ_n , but the two coverage requirements are compared at the same law for every n .

5.2 What the rate result does not show

The matching bounds are not constant-sharp and do not imply practically short confidence sets. Supplement Section S.3.1 gives a fixed population example in which compression to (Y, D, Z, S) hides a shift in the full-covariate target; that example is explanatory and does not replace the full-observation testing argument. Supplement Section S.3.2 evaluates the explicit finite-sample constants. At $\beta = 1/8$, $n = 10^6$, and $\lambda_n = 1/32$, the available upper bound is still the trivial diameter bound two. The exhibits are analytic calculations rather than Monte Carlo evidence or a claim of estimator dominance.

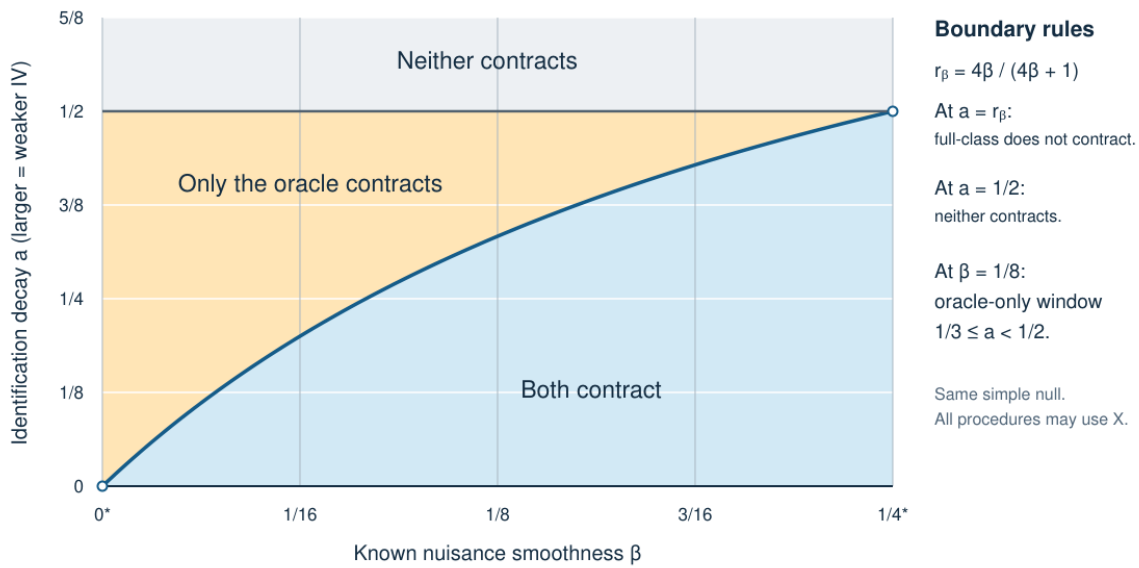
6 Discussion and Conclusion

Theorem 3.1 and Proposition 3.2 distinguish true nuisance simplicity from knowledge that the restriction is valid. At the reference law P_n^* , all three nuisance means are constant. Requiring coverage over the larger Hölder model nevertheless changes the optimal expected-diameter rate from $q_{0,n}$ to $q_{\beta,n}$. Both comparisons use the same observations and the same law; the oracle receives only the small-model restriction.

The IV-compatible alternatives are essential to this model-specific conclusion. They retain overlap, cell interiority, exclusion, conditional instrument independence, treatment monotonicity, and the identification stratum while shifting the causal ratio. The shared-sign mixture transfers

When does a simple IV model permit short confidence sets?

Identification path: $\lambda_n = \ell n^{-a}$, $\ell = 1/32$. Smoothness β is fixed and known.



Asymptotic optimal expected-diameter orders only; constants are suppressed. Not realized CI lengths.

* $\beta = 0$ and $\beta = 1/4$ are excluded. Source: Theorem 3.1, Proposition 3.2, and Section 3.4.

Figure 1: Contraction regimes under different honesty requirements. The comparison is at the same simple law $P^* = P_n^*$ from (3.1). Full-model honesty permits optimal expected-diameter contraction if and only if $a < r_\beta$, whereas oracle honesty permits contraction if and only if $a < 1/2$. All procedures may use the full covariate X . The endpoints $\beta = 0$ and $\beta = 1/4$ are excluded. Constants are suppressed; the display is analytic, not a simulation.

coverage requirements to the simple law, and the projection-and-inversion construction attains the rate over the same large model. The testing, adaptation, projection, and inversion tools have established precedents; the contribution is their connection through this valid IV construction and same-law comparison.

The result remains deliberately narrow. It assumes binary outcome, treatment, and instrument; a known uniform one-dimensional covariate distribution; a fixed two-cell summary; known fixed $\beta \in (0, 1/4)$; and the stated overlap, interiority, and identification bounds. The target is the overlap-weighted complier effect N/Λ , not an unweighted LATE or an ATE. Unknown covariate distributions, learned summaries, unknown smoothness, higher-dimensional covariates, the boundary $\beta = 1/4$, sharp constants, and empirical performance remain open.

Accordingly, the paper establishes a restricted theoretical comparison rather than a new general estimation method. Simple nuisance means alone do not justify the oracle confidence-set rate when coverage must remain valid over the larger class.

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